Distributed Real Time Control Systems

MEEC

Report – Project 1st Stage

Control of an illumination system model

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1 System Model

The luminaire consists of a white LED driven by a PWM signal ($u \in [0, 1]$) proportional to the analogue voltage applied to the PWM pin and a light-dependent resistor (LDR) measuring the reflected illuminance inside a shoebox model lined with white paper to maximize reflectivity, the interior can be seen in Fig. 1 The overall system is decomposed into a Plant and a Sensor block, as shown in Fig. 2.Assuming instantaneous LED and light propagation dynamics, the plant is the static affine model (Equation. 1), where $G\,[\mathrm{lx/dc}]$ is the system gain and $d(t)$ is the external illumination.

The sensor is a dynamic non-linear system, with a hardware implementation shown in Fig. 3, where the LDR forms a voltage divider with $R_1 = 10\,\mathrm{k}\Omega$ at $V_{cc} = 3.3\,\mathrm{V}$. The raw ADC reading $v_{\mathrm{ADC}}(t)$ is first converted to voltage $v(t)$ via Equation. 2 where $\mathrm{ADC}_{\mathrm{range}} = 4096$ (12-bit resolution). The voltage is then converted to resistance via Equation 3, and finally to illuminance through the log-log characteristic (Equation 4), with datasheet slope $m = -0.8$. The intercept b is calibrated by placing the sensor under the lab fluorescent lights at a known illuminance of $500\,\mathrm{lx}$, yielding $b = 6.3044$.

2 Controller Design

Since the system to control is 0th order (affine function), there is no need for a derivative term, a PI controller suffices to achieve zero steady-state error. Set-point weighting on the proportional term is included to decouple reference tracking from disturbance rejection, giving the controller two degrees of freedom (Equation. 5). The feedback path is tuned using k_p, k_i ($k_d = 0$) to ensure effective disturbance rejection, while the feedforward path uses the weighting parameters b ($c = 0$) to shape a fast and smooth response to set-point changes.The forward-Euler discretisation at sampling time $h = t_{k+1} - t_k = T_s = 10\,\mathrm{ms}$ ($100\,\mathrm{Hz}$) yields Equation. 6.

If the controller receives a reference that cannot be achieved due to the physical limits of the LED, the actuator saturates and a difference emerges between what the system can provide with $u(t)$ clamped between 0 and 1, and what the controller asks of the system $v(t)$. Since the measured lux will never reach the reference, the integral error will keep accumulating due to the constant tracking error until a feasible reference is provided. When the reference drops to a reachable value, the integrator is wound up, making the controller overshoot and respond slowly while the integrator unwinds. To solve this problem two anti-windup mechanism were compared:

- **Stop-integrating:** the update $i(t_{k+1}) = i(t_k) + k_i h\, e(t_k)$ is suppressed whenever $v \geq 1$ and $e > 0$, or $v \leq 0$ and $e < 0$. This guarantees zero integrator growth during saturation and instant, undershoot-free recovery when the reference returns to a feasible value (Fig. 5b).

- **Back-calculation:** an additional term $k_t h\,(u - v)$, where $e_s = (u - v)$ is added to the integrator update, providing smooth unwinding at the cost of an extra tuning parameter k_t. This creates a self limiting process, where as $k_i e$ grows $k_t e_s$ grows negatively, canceling each other out. Making v settle at a value just above the saturation border in steady state, responding immediately when conditions change and saturation is released (Fig. 5a).

A bumpless transfer mechanism (Equation. 7) is applied whenever controller gains k_p or b change at run-time, updating the integrator so that the controller output u is continuous across the parameter change.

A static feedforward term initializes the integrator to $u_{ff} = (r - d)/G$ at loop closure, placing the LED immediately near the correct operating point so the PI only corrects the small residual error. This seeding is applied only once at feedback enable, not on subsequent reference changes, to avoid overwriting the running integrator mid-flight.

3 Results

Beyond the standard command set specified in the project guide, a suite of calibration and tuning commands was added, all grouped under the prefix `c` (Table 1). The gain G and background illuminance d depend on luminaire placement and ambient conditions, so they are calibrated at the start of each session via `c g`, and linearity verified via `c s`. Figure 4 shows one such calibration, yielding $G = 45.894\,\text{lux/dc}$ and $d = -1.583\,\text{lux}$ ($R^2 = 0.9964$), confirming that $x(t) = G \cdot u(t) + d$ holds over the full actuator range.

To reduce LDR noise, the firmware implements a selectable digital filter via `c f` *`i`* *`mode`*: no filter (`0`), mean of 64 samples (`1`), or median of a 9-sample window (`2`). Figure 7 shows the mean filter performs best ($\approx 0.2\,\text{s}^{-1}$), as uniform averaging suppresses broadband noise; the median reaches $\approx 1.1\,\text{s}^{-1}$ since it passes slower noise fluctuations; without filtering flicker grows to $\approx 2.9\,\text{s}^{-1}$. Sampling jitter averages $\langle J \rangle = 0.91\,\mu\text{s}$ (0.0091% of h), confirming near-ideal periodicity despite concurrent tasks.

Sensor transient dynamics were characterised by stepping $u: \ 0 \to 1$ and $u: \ 1 \to 0$ with filtering disabled (Fig. 6, Eq. 9–10), yielding $\tau_{\text{rise}} = 615\,\text{ms}$ and $\tau_{\text{decay}} = 120\,\text{ms}$. The asymmetry follows from the coupled LDR–RC model: on the rise the capacitor charges through $R_{\text{LDR,dark}} \gg R_1$; on the decay it discharges through R_1 alone.

Fig. 8 compares two tuning cases: with $k_i = 0.3$ the faster integral action causes the duty cycle to saturate briefly at the transient, producing a small overshoot of approximately 2 lx; reducing k_i to 0.155 eliminates the overshoot while preserving a fast rise time and zero steady-state error.

To evaluate disturbance rejection, the box lid was repeatedly opened and closed while the reference was held at 25 lx. A strong external disturbance exceeding the reference was introduced, making it temporarily unreachable. As seen in Fig. 9, the controller drives the duty cycle to zero, and once the disturbance is removed, rapidly reconverges to the reference.

To evaluate the anti-windup mechanisms, the reference was stepped to an unreachable 70 lux, forcing sustained saturation at $u = 1$, before being stepped down to 5 lux. As shown in Fig. 10, without anti-windup the integrator winds up and delays the duty cycle response by $\approx 11\,\text{s}$. With stop-integrating, the duty cycle responds immediately with no windup delay. Back-calculation was also evaluated with the results shown in Fig. 11: higher k_t discharges the integrator faster, reducing undershoot and recovery time after the reference step down.

Bumpless transfer was verified by stepping $b: 1 \to 0 \to 1$ at steady state. Without it, a spike of $\Delta u \approx 0.7$ appears in the duty cycle at each gain change; with it the integrator absorbs the discontinuity and the transition is smooth, as shown in Fig. 12.

4 Attachments

4.1 Equations

Plant model: static affine relationship between duty cycle $u \in [0,1]$ and real luminance x, where $G\,[\mathrm{lx/dc}]$ is the system gain and $d(t)$ external illumination.

$$x(t) = G\,u(t) + d(t) \tag{1}$$

ADC Voltage Conversion,where $\mathrm{ADC}_{\mathrm{range}} = 4096$ corresponds to the 12-bit resolution of the RP2040.

$$v(t) = \frac{v_{\mathrm{ADC}}(t)}{\mathrm{ADC}_{\mathrm{range}}} \cdot V_{cc} \tag{2}$$

LDR Resistance. The voltage $v(t)$ is related to the LDR resistance R_{LDR} through the voltage divider with $R_1 = 10\,\mathrm{k}\Omega$ and $V_{cc} = 3.3\,\mathrm{V}$:

$$v(t) = \frac{R_1}{R_1 + R_{\mathrm{LDR}}} \cdot V_{cc} \quad \Rightarrow \quad R_{\mathrm{LDR}} = R_1 \left(\frac{V_{cc}}{v(t)} - 1 \right) \tag{3}$$

LUX conversion: converts the voltage $v(t)$ to illuminance y via the LDR resistance $R_{\mathrm{LDR}} = R_1(V_{cc}/v - 1)$ and the log-log characteristic, with slope $m = -0.8$ and calibrated intercept $b = 6.3044$.

$$\begin{cases} R_{\mathrm{LDR}} = R_0 x^{-\gamma} \\ \log_{10}(R_{\mathrm{LDR}}) = \log_{10}(R_0) - \gamma \log_{10}(x) \end{cases} \implies \begin{cases} \log_{10}(R_{\mathrm{LDR}}(x)) = m\,\log_{10}(y) + b \\ y = 10^{\frac{\log_{10}(R_{\mathrm{LDR}}(x)) - b}{m}} \end{cases}$$

$$m = -\gamma = -0.8, \qquad b = \log_{10}(R_0) = 6.3044 \tag{4}$$

Continuous-time PI with set-point weighting: the proportional term acts on the weighted error $br - y$ to reduce overshoot on reference steps, while the integral term acts on the full tracking error e.

$$\begin{aligned} e(t) &= r(t) - y(t) \\ u(t) &= k_p\Big[b\,r(t) - y(t)\Big] + \int_0^t k_i\,e(\tau)\,\mathrm{d}\tau \end{aligned} \tag{5}$$

Forward-Euler discretisation: the proportional part p and integral accumulator i are computed separately at sampling period $h = 10\,\mathrm{ms}$ to allow independent anti-windup clamping of i.

$$\begin{cases} e(t_k) = r(t_k) - y(t_k), \\ u(t_k) = p(t_k) + i(t_k) \end{cases} \quad \Rightarrow \quad \begin{cases} p(t_k) = k_p\Big[b\,r(t_k) - y(t_k)\Big], \\ i(t_{k+1}) = i(t_k) + k_i\,h\,e(t_k) \end{cases} \tag{6}$$

Bumpless transfer: correction applied to the integrator state when gains are updated online, ensuring continuity of the control output u at the moment of the change.

$$i \leftarrow i + k_{p,\mathrm{old}}\Big[b_{\mathrm{old}}\,r - y\Big] - k_{p,\mathrm{new}}\Big[b_{\mathrm{new}}\,r - y\Big] \tag{7}$$

LED maximum power: estimated from the actual circuit operating point, where $R_2 = 47\,\Omega$ is the current-limiting resistor and $V_F = 3.1\,\mathrm{V}$ is the typical forward voltage from the LED datasheet.

$$P_{\max} = V_F \cdot I_F, \qquad I_F = \frac{V_{cc} - V_F}{R_2} = \frac{3.3 - 3.1}{47} \approx 4.26\,\mathrm{mA} \tag{8}$$

Sensor transient dynamics: the sensor output is governed by two coupled first-order differential equations. The LDR resistance R_{LDR} chases its steady-state value $R_0\, x^{-\gamma}$ with an illuminance-dependent time constant $\tau_{\text{LDR}}(x)$; the capacitor voltage v then charges or discharges through the voltage divider formed by $R_{\text{LDR}}(t)$ and R_1:

$$\begin{aligned} \tau_{\text{LDR}}(x)\,\frac{dR_{\text{LDR}}(t)}{dt} + R_{\text{LDR}}(t) &= R_0\, x(t)^{-\gamma} \\ C\,\frac{dv(t)}{dt} + \frac{v(t)}{R_1} &= \frac{V_{cc} - v(t)}{R_{\text{LDR}}(t)} \end{aligned} \tag{9}$$

The two equations are coupled in series: $x(t)$ drives $R_{\text{LDR}}(t)$ through the first equation, which in turn drives $v(t)$ through the second. The measured time constants therefore reflect both contributions:

$$\tau_{\text{rise}} \approx \tau_{\text{LDR}}(x) + R_{\text{LDR,dark}}\, C, \qquad \tau_{\text{decay}} \approx \tau_{\text{LDR}}(x) + R_1\, C \tag{10}$$

On the rise ($u:\ 0 \to 1$), the capacitor charges through the large dark-state resistance $R_{\text{LDR,dark}} \gg R_1$, giving a slow effective τ_{RC}; experimentally $\tau_{\text{rise}} = 615\,\text{ms}$. On the decay ($u:\ 1 \to 0$), R_{LDR} jumps to a very large value and the right-hand side of the second equation vanishes, so the capacitor discharges through R_1 alone, giving $\tau_{\text{decay}} \approx \tau_{\text{LDR,decay}} + R_1 C = 30 + 100 = 130\,\text{ms}$, where $\tau_{\text{LDR,decay}} = 30\,\text{ms}$ is taken from the PGM5659D datasheet. This is confirmed experimentally at $\tau_{\text{decay}} = 120\,\text{ms} \approx 130\,\text{ms}$, validating both the RC model and the datasheet value.

4.2 Figures

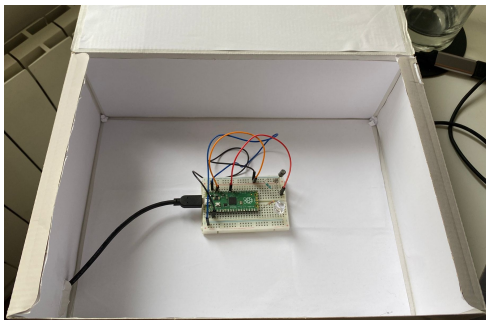


Figure 1: Model of an open office space with model luminaire.

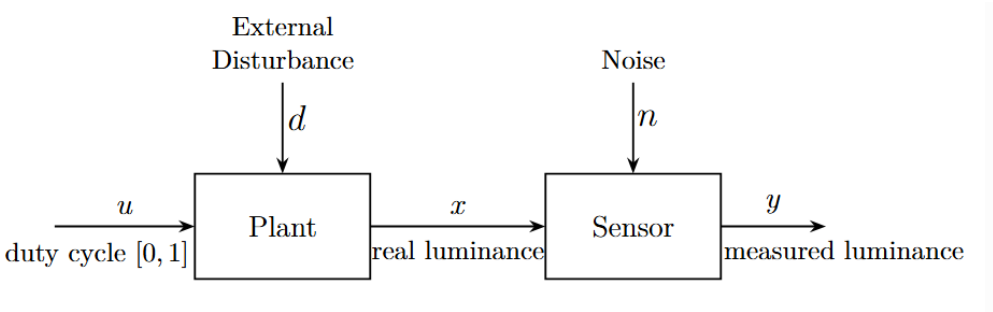


Figure 2: Block diagram of the luminaire model. The Plant maps duty cycle $u \in [0, 1]$ to real luminance x, subject to external disturbance d. The Sensor converts x to measured luminance y, corrupted by noise n.

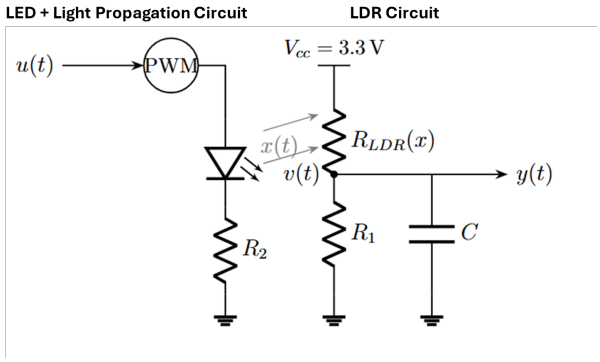


Figure 3: Luminaire electronics. Left: LED driving circuit (PWM signal through current-limiting resistor R_2). Right: illuminance reading circuit (R_{LDR}–R_1 voltage divider with capacitor C for noise filtering).

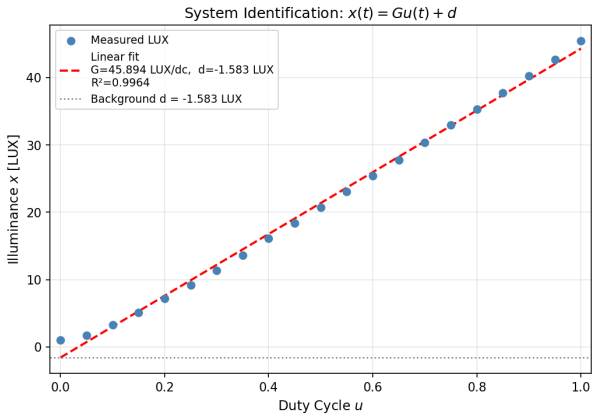
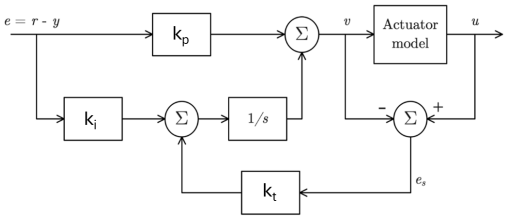
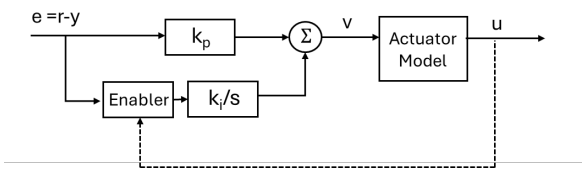


Figure 4: Static gain identification of the luminaire. Measured steady-state illuminance x as a function of duty cycle u, with linear fit $x(t) = G \cdot u(t) + d$, confirming a highly linear static response.



(a) Back-calculation anti-windup. The saturation error $e_s = u - v$ is fed back through gain k_t to unwind the integrator, where v is the unsaturated and u the saturated control signal.



(b) Stop-integrating anti-windup. The Enabler block gates the integrator update: integration is frozen whenever $(u \geq 1 \wedge e > 0)$ or $(u \leq 0 \wedge e < 0)$, preventing accumulation during saturation. The dashed line shows the u feedback path into the Enabler.

Figure 5: PI controller block diagrams for both anti-windup strategies.

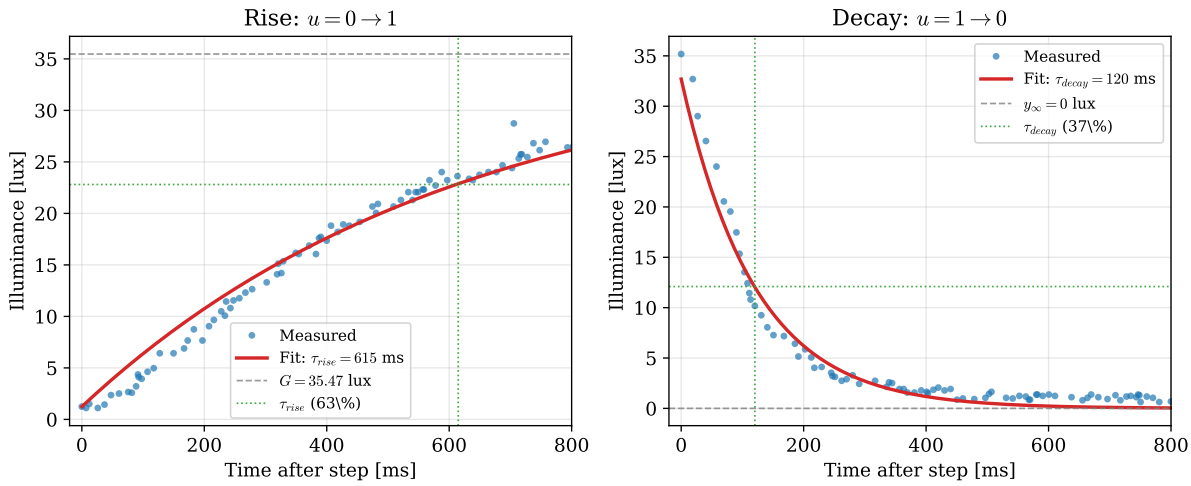


Figure 6: Measured sensor step response (digital filter disabled), yielding $\tau_{\text{rise}} = 615\,\text{ms}$ and $\tau_{\text{decay}} = 120\,\text{ms}$.

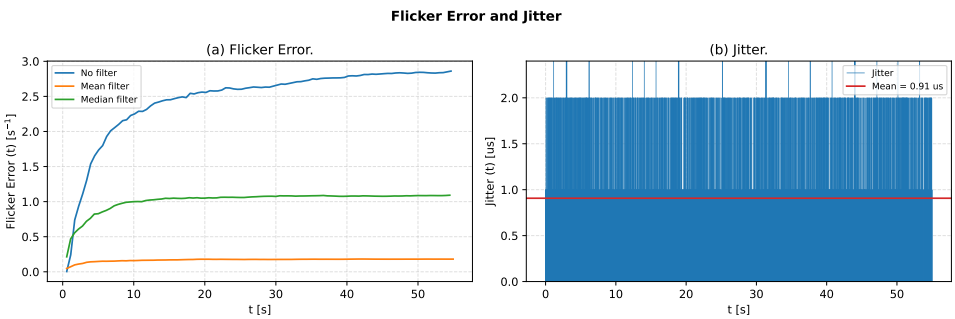
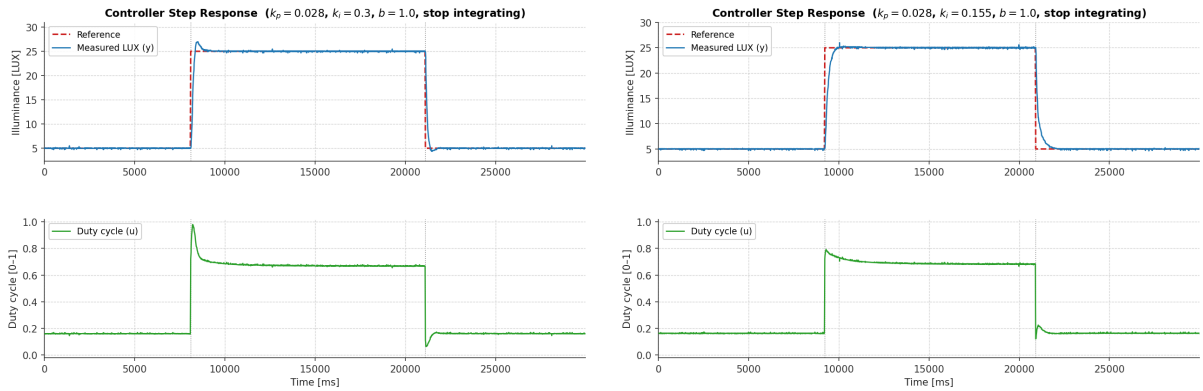


Figure 7: Flicker error and jitter comparison for median, mean and no digital filter



(a) Step response with $k_i = 0.3$ (overshoot)

(b) Step response after tuning ($k_i = 0.155$)

Figure 8: Effect of integral gain on closed-loop step response: higher k_i causes overshoot, lower k_i gives clean settling.

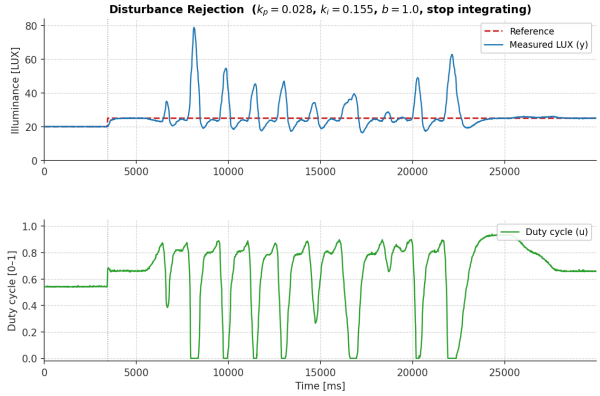
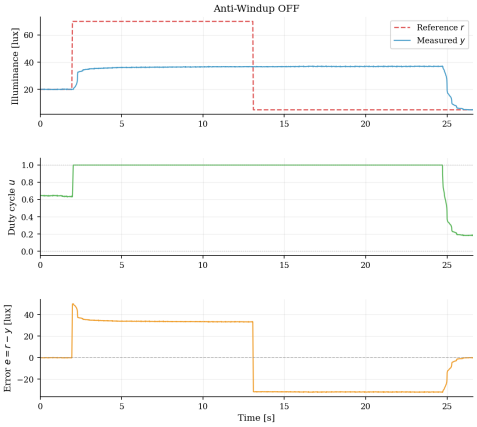
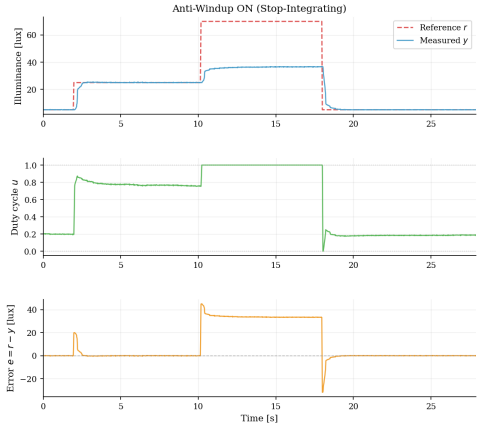


Figure 9: Result of opening and closing the box

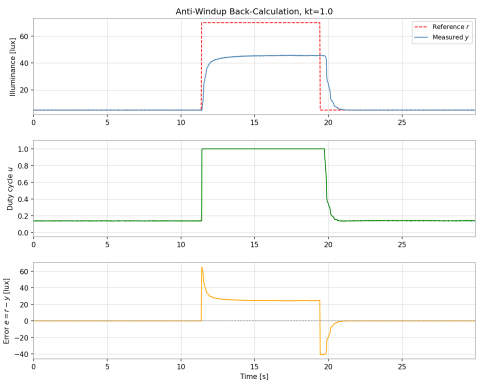


(a) Without anti-windup: integrator winds up, duty cycle stays saturated at $u = 1$ for $\approx 11\,\mathrm{s}$ after stepping down to $r = 5\,\mathrm{lux}$.

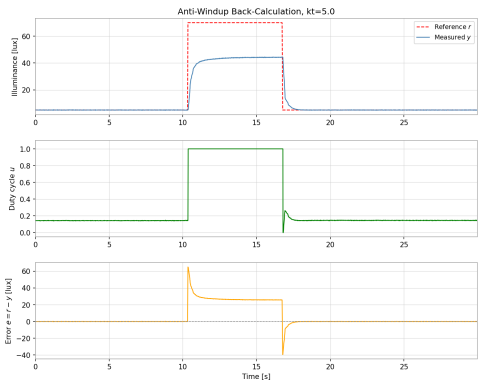


(b) With stop-integrating anti-windup: integrator is frozen during saturation, duty cycle responds immediately upon reference step down.

Figure 10: PI controller with and without anti-windup under reference sequence $20 \to 70 \to 5\,\mathrm{lux}$. The $70\,\mathrm{lux}$ target forces sustained output saturation ($G \approx 45\,\mathrm{lux/dc}$).

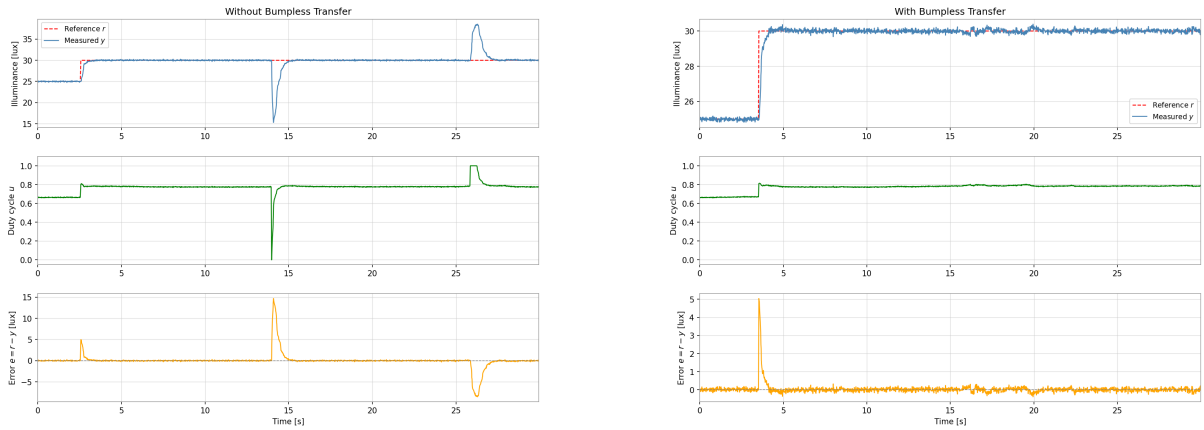


(a) $k_t = 1.0$



(b) $k_t = 5.0$

Figure 11: Back-calculation anti-windup under $r = 70 \to 5\,\mathrm{lux}$. Higher k_t accelerates integrator discharge, reducing undershoot and recovery time.



(a) Without bumpless transfer

(b) With bumpless transfer

Figure 12: Effect of bumpless transfer on the control signal during a setpoint weight change ($b : 1 \to 0 \to 1$). Without bumpless transfer, the proportional term causes an abrupt jump in the duty cycle at each gain change; with bumpless transfer, the integrator absorbs the discontinuity and the control signal transitions smoothly.

4.3 Tables

Table 1: Additional commands implemented beyond the project guide specification.

Command	**Response**	**Description**
Calibration (prefix `c`*)*		
`c b [val]`	`ack` / LUX	Sets b manually and immediately prints the resulting LUX reading. Omitting `val` prints the current reading without changing b.
`c m <val>`	`ack`	Sets the LDR log-log slope m (default -0.8 from datasheet).
`c g`	`ack`	Calibrates system gain G and background illuminance d by toggling the LED off then fully on inside the box.
`c s [steps]`	CSV + `END`	Performs a duty-cycle sweep $0 \to 1$ in `steps` increments (default 10)
`c p <i> <val>`	`ack`	Sets proportional gain k_p of luminaire i at runtime.
`c i <i> <val>`	`ack`	Sets integral gain k_i of luminaire i at runtime.
`c B <i> <val>`	`ack`	Sets set-point weight b
`c t <i> <val>`	`ack`	Sets back-calculation gain k_t for anti-windup mode 1.

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Command	**Response**	**Description**
`c w <i> <0|1>`	`ack`	Enables (1) or disables (0) bumpless transfer on runtime gain changes.
`c f <i> <0|1|2>`	`ack`	Sets LDR filter mode: 0 = none, 1 = mean, 2 = median.
`c ?`	param list	Prints all current calibration and controller parameters: m, b, G, d, k_p, k_i, b (set-point weighting), k_t, anti-windup mode.
Control		
`a <i> <0|1|2>`	`ack`	Sets anti-windup mode: 0 = off, 1 = back-calculation, 2 = stop-integrating.
`R`	`ack`	Resets all performance metrics (energy, visibility error, flicker error) to zero.
Stream		
`s j <i>`	stream	Streams the control loop jitter [μs], i.e. the deviation of the sampling period from the nominal 10 ms.
`S j <i>`	`ack`	Stops the jitter stream.
Help		
`h`	command list	Prints the full command reference to the serial monitor.